

5 The Hamiltonian

The Hamiltonian is conserved if $\frac{\partial \mathcal{L}}{\partial t} = 0$.

$$H = \left(\sum_{i=0}^{\infty} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \dot{q}_i \right) - \mathcal{L}$$

The Hamiltonian is not always just the mechanical energy - if the system is driven (i.e. some external constraint).

		Is the coordinate transformation explicitly time dependent?	
Is $\mathcal{L}$ explicitly time dependent?		Yes	no
	Yes	H is not conserved $H \neq T + U$	H is not conserved $H \neq T + U$
	No	H is conserved $H = T + U$	H is conserved $H = T + U$

For example: The bead on a rotating hoop (of constant azimuthal rate ω_0) wasn't explicitly time dependent, but because it was driven, the theta vector pointed differently in space over time. So H is conserved, but $H = T + U - mg \sin^2 \theta x^2$ which is not $T + U$.